

WHY APÉRY IS ALONE: A SCAN OF THE MODULAR IRRATIONALITY RACE

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ABSTRACT. Apéry’s 1978 proof that $\zeta(3) \notin \mathbb{Q}$ is usually described as a miracle: a recurrence appears, its two solutions have the right denominators, and a race between denominator growth and approximation decay is won by a hair. We recall the race in a form that makes its two competing exponents visible, and then — following a scanner proposal of our collaborator Sol — we run the race over an entire class of constructions: the companions attached to genus-zero modular curves with an Atkin–Lehner fold.

Two things come out. First, a unification: the classical Apéry $\zeta(3)$ source itself *vanishes at the fold*, $\Phi_\gamma(\tau_c) = 0$ [VERIFIED: 10^{-122}], because it is the W_6 -odd Eisenstein direction. Apéry’s construction is therefore an instance of exactly the mechanism used to manufacture cuspidal companions — an odd source vanishing at an Atkin–Lehner fixed point — realised in the Eisenstein class at a level whose fold ratio happens to exceed e^3 . Second, a bounded negative: over standard integral eta-quotient hauptmoduln of level $N \leq 25$, level 6 is the *unique* winner of the $r = 3$ race, and its odd cusp space $S_4^-(\Gamma_0(6))$ is zero; every level carrying odd cusp forms loses the race by a factor ≥ 4 . The boundary levels 10 and 14, which sit at score exactly e^3 , are settled by direct computation as well. Along the way the scan produces new arithmetic: the odd-cusp companion at level 10 has denominator exponent $\kappa = 3$ at generic primes, with deflation only at level primes, and an exact 5-adic descent law $v_5(B_{5m}) = v_5(B_m) - 2$ [VERIFIED: $m \leq 11$]. We state what is proved, what is only computed, and the two problems that survive.

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1. INTRODUCTION: THE RACE

1.1. Apéry's theorem, in the shape we need. In 1978 Apéry announced that $\zeta(3)$ is irrational. The proof, as digested by van der Poorten, Cohen and Beukers, fits on one page once one knows the two sequences. Put

$$A_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2,$$

the *Apéry numbers* $1, 5, 73, 1445, \dots$, and let B_n be the second solution of the same three-term recurrence

$$n^3 u_n = (34n^3 - 51n^2 + 27n - 5) u_{n-1} - (n-1)^3 u_{n-2}, \quad (1)$$

normalised by $B_0 = 0, B_1 = 6$. Then $A_n \in \mathbb{Z}$, the rational numbers B_n have denominators dividing $2d_n^3$ where $d_n = \text{lcm}(1, \dots, n)$, and

$$\frac{B_n}{A_n} \longrightarrow \zeta(3).$$

So far this is only a fast approximation scheme. The proof is the bookkeeping that follows. Three exponential rates govern everything:

- the *growth rate* α : $A_n \sim \alpha^n$ up to subexponential factors, here $\alpha = (1 + \sqrt{2})^4 \approx 33.97$;
- the *decay rate* β : $|A_n \zeta(3) - B_n| \sim \beta^n$, here $\beta = (\sqrt{2} - 1)^4 \approx 0.0294 = 1/\alpha$;
- the *denominator rate* δ : a common denominator for A_n, B_n grows like δ^n , here $\delta = e^3$ because $d_n \sim e^n$ by the prime number theorem and the exponent is 3.

Clearing denominators produces genuine integers $\tilde{A}_n = d_n^3 A_n$ and $\tilde{B}_n = 2d_n^3 B_n$ with

$$0 < |\tilde{A}_n \zeta(3) - \tilde{B}_n| \approx (\delta \beta)^n.$$

If $\zeta(3) = p/q$ were rational, the left-hand side would be a nonzero integer multiple of $1/q$, hence bounded below by $1/q$; but the right-hand side tends to 0 as soon as $\delta \beta < 1$. Contradiction. Everything therefore rests on a single inequality.

Definition 1.1 (The race condition; Sol's score). For a construction with rates (α, β, δ) as above, the *score* is

$$\text{score} := \delta |\beta|,$$

and the construction *wins the race* (i.e. proves irrationality of its limit) precisely when $\text{score} < 1$.

For Apéry’s $\zeta(3)$: $\delta|\beta| = e^3 \cdot 0.0294 \approx 20.09/33.97 \approx 0.59 < 1$. The margin is 41%; there is no slack to spare and none of the three rates could be much worse. For Apéry’s $\zeta(2)$ construction (the $\binom{n}{k}^2 \binom{n+k}{k}$ family) the corresponding numbers are $\delta = e^2$, $|\beta|^{-1} = \varphi^{10} \approx 11.09$, and the score is 0.67.

Remark 1.2. The formulation of the race as a one-number *score* attached to a construction — and the proposal to scan it systematically over a class of constructions rather than admire it in the two classical cases — is due to our collaborator Sol. The entire paper is an execution of that proposal. We keep the terminology.

1.2. Why the score is computable in the modular world. The reason a scan is possible at all is that for the constructions we care about — the ones attached to modular curves — the three rates are not independent mysteries. They are read off from one picture.

Suppose Γ is a genus-zero congruence group with hauptmodul t (a generator of its function field, normalised to $t = q + O(q^2)$), and suppose F is a weight- k modular form on Γ that, written as a function of t , satisfies a Picard–Fuchs equation $LF = 0$ whose coefficient sequence $A_n = [t^n]F$ is integral. In the sporadic cases this is exactly the classical dictionary between Apéry-like sequences and modular parametrisations. The differential operator L has singularities at the roots of a polynomial $P(t)$; write t_c for the root nearest the origin (the *fold*) and t'_c for the conjugate root. Then:

- $\alpha = 1/|t_c|$ — the growth of A_n is governed by the nearest singularity;
- if the companion B_n is built from a *source* that vanishes at the fold, the particular solution’s own singularity is pushed out to t'_c , so the error decays like $|t'_c|^{-n}$, i.e. $|\beta| = 1/|t'_c|$;
- $\delta = e^r$ where r is the denominator exponent: $r = 3$ for a weight-4 source over a weight-2 family, $r = 2$ for weight-3 over weight-1.

Hence

$$\boxed{\text{score}_r = \frac{e^r}{|t'_c|}} \tag{2}$$

and the race is won exactly when the conjugate singularity of the Picard–Fuchs operator is farther from the origin than e^r . That is a statement about one complex number attached to a level, and one can enumerate levels.

For Apéry’s $\zeta(3)$ the relevant curve is level 6 with $P(t) = 1 - 34t + t^2$, whose roots are $t_c = 17 - 12\sqrt{2} \approx 0.0294$ and $t'_c = 17 + 12\sqrt{2} \approx 33.97$, giving score $20.09/33.97 = 0.59$ as above. The whole of §3 is the systematic version of this one line.

1.3. What this paper contains. §2 explains the structural discovery that makes the scan meaningful: Apéry’s source vanishes at the fold, so his construction is the *same* construction as the deliberately manufactured cuspidal apparatus of the companion paper — one mechanism in two representation-theoretic classes. §3 runs the scan and reports the verdict, together with a lesson about normalisation dependence that changes what the optimisation problem even is. §4 settles the two boundary levels. §5 reports the arithmetic that the scan turned up as a by-product — a prime-local denominator law and an exact 5-adic descent law — and closes the “arithmetic loophole”. §6 states honestly what is bounded, assumed, and unproved.

The intended reader is a graduate student who has seen Apéry’s proof once and knows what a modular form is. We have kept the modular prerequisites to: hauptmoduln, Atkin–Lehner involutions, oldforms, and the dimension formula for $S_k(\Gamma_0(N))$.

1.4. Evidence discipline. This programme labels every assertion. [PROVED] means a paper proof exists; [THM] means proved and adversarially refereed; [VERIFIED: range] means exact computation

over a stated finite range and is *never* treated as proof; [EXCLUDED] means a proved negative; [OPEN] means open, stated in its sharpest known form. Nothing in this paper is asserted at a higher grade than in its source ledger. In particular the central negative result of §3 is a bounded computational verdict over an explicitly delimited search domain, not a theorem about all modular constructions — and we say so again in §6.

2. THE UNIFICATION: APÉRY’S SOURCE VANISHES AT THE FOLD

2.1. Sources and companions. Fix a genus-zero level N , a hauptmodul t , and a weight-2 modular form F with $A_n = [t^n]F \in \mathbb{Z}$ and Picard–Fuchs operator L . A *companion* of F is a second solution: a sequence B_n satisfying the same recurrence with an inhomogeneity. The standard way to produce one is to pick a modular form g of weight $r + 1$ (the *source*) and set

$$B = F \theta_q^{-r} g, \quad \theta_q = q \frac{d}{dq},$$

where θ_q^{-r} is the r -fold formal antiderivative in q (an Eichler integral). Expanding B in powers of t gives B_n , and the operator applied to B returns a computable inhomogeneity: in the normalised form used throughout this programme,

$$L(F \theta_q^{-r} g) = \Phi \quad \text{with} \quad \Phi = \frac{t \sigma^r}{P F} \cdot (\dots),$$

where $\sigma = \theta_q t/t$. The point of the modular-anchor programme is that Φ is again a modular object, and that for all twelve identifiable sporadic families it is a *pure Eisenstein series* of weight $r + 1$ [VERIFIED: q^{60} , held-out band q^{27} – q^{60}].

Apéry’s B_n is the case $N = 6$, $r = 3$, with source Φ_γ the weight-4 Eisenstein direction on $\Gamma_0(6)$.

2.2. The vanishing mechanism, first seen in the cuspidal case. The mechanism was isolated in the reverse direction. On Domb’s curve (level 12, $\dim S_4(\Gamma_0(12)) = 2$, spanned by the two embeddings $f_6(q), f_6(q^2)$ of the level-6 newform $f_6 = (\eta_1 \eta_2 \eta_3 \eta_6)^2$), one has the Atkin–Lehner action

$$(f_6|_4 W_{12})(\tau) = 4f_6(2\tau) \quad [\text{PROVED}],$$

which makes

$$f^* = f_6(q) - 4f_6(q^2) = q - 6q^2 - 3q^3 + 12q^4 + 6q^5 + \dots \in S_4(\Gamma_0(12))$$

the W_{12} -odd eigenvector. Because the weight-4 automorphy factor at the fixed point $\tau_{12} = i/\sqrt{12}$ equals 1, oddness forces $f^*(\tau_{12}) = -f^*(\tau_{12})$, i.e.

$$f^*(\tau_{12}) = 0.$$

That single vanishing does two independent jobs (see the companion paper on the cuspidal apparatus, and [8, §4]):

- (i) **Exponential convergence.** The inhomogeneity $t f^*/\Phi$ becomes regular at the fold, so the particular solution is singular only at the *conjugate* root; the error ratio is $(t_c/t'_c)^n$ instead of $O(1)$.
- (ii) **Clean limit.** For an $\varepsilon = -1$ source, the fixed-point Eichler equation selects value-plus-quasiderivative data that the odd period polynomial reduces to a single critical L -value; the second-kind (quasiperiod) term is structurally absent.

The resulting apparatus has $d_n^3 B_n^* \in \mathbb{Z}$ [VERIFIED: $n \leq 40$], error $\approx 4^{-n}$, and $\lim B_n^*/A_n = L(f_6, 3)/2$ [VERIFIED: PSLQ, 61 digits]. Its score, however, is $e^3/4 \approx 5.0 > 1$: it does *not* prove irrationality of $L(f_6, 3)$. It was built to test whether cuspidal L -values could be brought inside the machinery at all. They can.

2.3. The discovery. Having isolated “odd source vanishing at the Atkin–Lehner fold” as the mechanism, one naturally asks what Apéry’s own source does at his fold. The answer:

Finding 2.1 ($\Phi_\gamma(\tau_c) = 0$). For the level-6 family γ (Apéry’s $\zeta(3)$ construction), the source Φ_γ vanishes at the Atkin–Lehner fixed point $\tau_c = i/\sqrt{6}$:

$$\Phi_\gamma(\tau_c) = 0 \quad [\text{VERIFIED: } |\Phi_\gamma(\tau_c)| < 10^{-122}].$$

The reason is the same parity argument as for f^* : Φ_γ lies in the W_6 -odd part of the weight-4 space, and any W_N -odd weight-4 form vanishes at the fixed point because the automorphy factor there is $+1$.

We label the numerical statement [VERIFIED] and the parity explanation [PROVED] (it is the two-line argument just given, once the eigenvalue $\varepsilon(\Phi_\gamma, W_6) = -1$ is known; the eigenvalue itself is read off the Eisenstein decomposition).

The consequence is conceptual, and it is the reason this paper exists:

Observation 2.2 (One mechanism, two classes). Apéry’s $\zeta(3)$ construction is *not* a separate miracle from the manufactured cuspidal apparatus. Both are instances of:

an odd source, vanishing at an Atkin–Lehner-fixed fold, integrated against a modular family.

They differ only in which part of the weight- $(r+1)$ space the source lives in — Eisenstein or cuspidal — and in whether the level’s fold ratio beats e^r .

This organises the landscape into a 2×2 table.

	odd Eisenstein source	odd cusp source
level 6 ($ t'_c = 33.97 > e^3$)	Apéry’s $\zeta(3)$ proof — score 0.59, wins	none exists: $S_4^-(\Gamma_0(6)) = 0$ (f_6 is Fricke- <i>even</i>)
level 12 ($ t'_c = 4 < e^3$)	a $\zeta(3)$ -type companion; loses the race	the $L(f_6, 3)/2$ apparatus; loses the race (score ≈ 5.0)

TABLE 1. The mechanism in its two classes at the two relevant levels.

Reading Table 1 the obvious question writes itself.

Problem 2.3 (The cuspidal-Apéry cell). Is there a genus-zero group Γ , with a rectified family whose Atkin–Lehner-fixed fold has $|t'_c| > e^r$, and with $S_{r+1}^-(\Gamma) \neq 0$? Such a level would give a race-winning apparatus whose limit is a critical L -value of a cusp form — an irrationality proof for a cuspidal period.

Both diagonal cells of Table 1 are occupied; the top-right cell is empty for a reason specific to level 6 ($\dim S_4 = 1$ and that one form has the wrong parity). Is the emptiness an accident of small level? §3 answers.

Remark 2.4 (Two repairs that do not work). Two obvious fixes are excluded before the scan starts. (i) At level 6 one cannot subtract an Eisenstein multiple to force a cusp form to vanish at the fold, because Φ_γ already vanishes there — the correction term one would want is identically zero on the fold [VERIFIED]. (ii) Moving to a level with a complex fold does not help: the regularisation argument requires a *real* fold at an Atkin–Lehner fixed point, as the δ -family control test showed (there the fold is at $q_c \approx 0.1137 - 0.1970i$, f^* does not vanish, and the companion’s coefficient ratios oscillate with both conjugate folds singular).

2.4. Why the modular class is the right arena. A reader might object that we are scanning a small pond. Two control experiments justify the choice of arena; both are negative results about the *hypergeometric* alternative.

Zudilin's Catalan recurrence (a well-poised ${}_6F_5$ -world construction with limit Catalan's constant G) satisfies the universal boundary-defect structure of our formalism — $L(y_v) = \frac{13}{2}t$ exactly [VERIFIED: q^{40}] — but fails both halves of the modular-anchor mechanism: its canonical nome admits *no* integralising rational rescale (exhaustive scan over $\mu = \pm 2^a 3^b 5^c$, $|a| \leq 8$, $|b| \leq 3$, $|c| \leq 2$), and its fold connection value misses $(2/13)G$ by 3.2×10^{-2} , far above truncation error. The diagnosis is structural: the raw operator has order six with half-integer indicial pairs $\{0, 0, \frac{1}{2}, \frac{1}{2}, \cdot, \cdot\}$. [EXCLUDED: bounded]

Koutschan–Zudilin's elliptic L -value construction — the unique one in the literature realising elliptic L -values as Apéry limits — carries the same half-integer structure ($x^{n-1/2}$ integrands, $(2n \pm 1)$ coefficients). Executed at $z = 1/2$ (the conductor-32 CM case, where CM might have saved it), modular rectification fails maximally: canonical-nome denominators $1352, 1.7 \times 10^{13}, 1.9 \times 10^{25}$ at $n = 2, 3, 4$, no integralising rescale, and the indicial polynomial $\theta(\theta + 1)^2(2\theta - 1)^2(2\theta + 1)^2 \dots$ has exponent 0 only *simple*, so no Frobenius log-partner exists and the canonical coordinate is not even well-founded. [EXCLUDED: bounded]

Lesson 2.5. The source of a companion depends on the *geometric realisation*, not on the constant. The same G is reached inside the modular class (sporadic family E, level 8, pure weight-3 χ_{-4} -Eisenstein source, integral mirror map) and outside it (Zudilin's recurrence, no modular structure in the canonical coordinate). Denominator and congruence phenomena belong to the realisation.

Consequently: the known non-modular L -value recurrences are hypergeometric-realised and outside our mechanism, while the mechanism itself has exactly one known race winner per weight. Scanning the modular class is scanning the place where the winners live.

3. THE SCAN

3.1. Method. The scanner (`work/z5eps/eps65_scanner.py`) implements (2) directly. For each level N in a list of levels carrying a standard integral eta-quotient *hauptmodul* t , it computes:

- Step 1. The *hauptmodul*.** $t = \prod_m \eta(m\tau)^{e_m}$ with the standard exponent vectors — e.g. level 6: $(\eta_1\eta_6/\eta_2\eta_3)^{12}$; level 12: $(\eta_1\eta_{12}/\eta_3\eta_4)^4$; level 10: $(\eta_1\eta_{10}/\eta_2\eta_5)^6$; prime levels $(\eta_1/\eta_N)^{24/(N-1)}$.
- Step 2. The *Atkin–Lehner Möbius map*.** $t \circ W_N$ is again a *hauptmodul*, hence $t \circ W_N = (at + b)/(ct + d)$ for constants determined by the level. These are fitted numerically from samples $\tau = iy$, $y \in \{0.35, 0.45, 0.55, 0.65\}$, with the fourth sample used as a consistency check (residuals reported per level).
- Step 3. The *critical values*.** The singular values of the operator are the critical values of t , i.e. the values of t at the zeros of

$$\sigma/t = \frac{1}{24} \sum_m e_m m E_2(q^m),$$

found by seeded Newton iteration over the complex disc — both the real folds and the complex second classes. Equivalently, and as a cross-check, they are the fixed points of the Möbius involution of Step 2: solving $ct^2 + (d - a)t - b = 0$ gives $\{t_c, t'_c\}$, and t_c is identified as the one nearest $t(i/\sqrt{N})$.

- Step 4. The *score and the cusp dimension*.** Output $\text{score}_r = e^r/|t'_c|$ for $r = 2, 3$, together with $\dim S_4(\Gamma_0(N))$ from the classical genus/elliptic-point formula.

3.2. Validation anchors. A scan is worth only its anchors. The following were reproduced *exactly* by the pipeline before any new level was trusted:

- **Level 6** (Apéry's $\zeta(3)$): $t_c = 0.0294\dots$, $|t'_c| = 33.97\dots$ — the roots of $1 - 34t + t^2$, known from the family operator.
- **Level 12** (Domb): $t_c = 1/16$, $|t'_c| = 1/4$ — the roots of $P_\alpha = (1 - 4t)(1 - 16t)$.
- **Level 8** (family ε): $(0.0429\dots, 1.457\dots)$.
- **Level 20** (family η): the complex fold $0.088 \pm 0.016i$, matching the known complex-fold classification (disc -16).
- **Level 5**: the symmetric $\pm 1/\sqrt{5}$ -scale pair, i.e. $|t_c| = |t'_c|$, consistent with $t \mapsto \kappa/t$.

3.3. **The table.** Table 2 is the full first run, at $r = 3$ ($\text{score}_3 = e^3/|t'_c| \approx 20.09/|t'_c|$; win iff < 1).

N	$\dim S_4$	t_c	$ t'_c $	score_3	verdict
2, 3, 4, 5,	0–5	$\pm\sqrt{\kappa}$	$= t_c $	—	no analytic gain
7, 9, 13, 25		symmetric			($t \mapsto \kappa/t$)
6	1	0.0294	33.97	0.59	WIN — but $S_4^-(\Gamma_0(6)) = 0$
8	1	0.0429	1.457	13.8	fail
10	3	0.0557	1.000	20.09	fail (exactly e^3)
12	3	0.0625	0.25	80	fail (Domb; the $L(f_6, 3)/2$ apparatus)
14	4	0.1497	1.000	20.09	fail (exactly e^3)
15	4	0.2011	0.726	27.7	fail
18	5	0.0895	$\lesssim 1$	$\gg 1$	fail (second class not fully resolved)
20	6	complex fold	—	—	excluded (no real AL fold)
24	8	0.0981	0.545	36.8	fail

TABLE 2. The $r = 3$ scan over standard integral eta hauptmoduln, $N \leq 25$.
[VERIFIED: numeric, 50 digits working precision]

The $r = 2$ scan ($\text{score } e^2/|t'_c|$, relevant for weight-3 sources over weight-1 families) produced no winner either, subject to the normalisation caveat of §3.6.

3.4. The verdict.

Finding 3.1 (The cuspidal-Apéry cell is empty in the scanned domain). Within the natural search domain — standard integral eta hauptmoduln, levels $N \leq 25$ — Problem 2.3 has no solution. Level 6 is the unique $r = 3$ race winner, and its odd cusp space vanishes; every level carrying odd cusp forms loses the race by a factor ≥ 4 . [VERIFIED: Table 2; domain as stated]

Or, in one sentence: *Apéry's $\zeta(3)$ apparatus is the unique odd-vanishing race-winning construction in its class — and it is Eisenstein because it has to be.* The classical picture, in which the Eisenstein-ness of Apéry's source looks like a lucky feature, inverts: the cuspidal alternative is not available at any level where the race can be won.

The reader should note carefully what kind of statement Finding 3.1 is. It is a verdict over an enumerated finite domain, computed numerically. It is *not* a theorem that no race-winning cuspidal construction exists. §6 lists precisely which doors it leaves open, and §3.6 explains why one of those doors is much wider than it first appears.

3.5. **Two boundary curiosities.** Levels 10 and 14 sit at $|t'_c| = 1.000$ to all computed digits — their conjugate class lies on the unit t -circle — hence at score *exactly* e^3 . This is the knife edge: the analysis contributes nothing either way, and the outcome is decided entirely by arithmetic. Since $\dim S_4 = 3$ and 4 there, odd cusp vectors could exist. Any denominator improvement whatsoever

— a $d_n^{3-\varepsilon}$ law of the sort the sporadic “deflation” phenomena hint at — would tip these levels into the win column and produce a cuspidal irrationality proof. This was the sharpest target the first run produced, recorded as:

Problem 3.2 (S-1). Determine the exact odd-cusp dimensions at levels 10 and 14, and whether any integral normalisation or auxiliary-source choice achieves denominator growth strictly below e^{3n} .

§4 answers it.

3.6. The normalisation lesson. Before that, a lesson that reshapes the problem. Look again at the first row of Table 2: at prime level the standard hauptmodul is *symmetric*, $t \circ W_N = \kappa/t$, so its two fixed points are $\pm\sqrt{\kappa}$ and $|t_c| = |t'_c|$ identically. There is zero analytic gain *by construction*. Yet the classical Apéry $\zeta(2)$ proof lives at level 5 and does win its race.

The resolution: the $\zeta(2)$ win exists in the $\Gamma_1(5)$ (asymmetric) normalisation of the level-5 hauptmodul, where $|t'_c|/|t_c| = \varphi^{10} \approx 123$; in the symmetric eta normalisation the same level shows no gain at all. Both are integral coordinates on the same curve.

Lesson 3.3 (Normalisation non-invariance). The race score is *not* an invariant of the level. It is an invariant of the *integral normalisation of the hauptmodul*. Two integral coordinates on the same modular curve can give scores on opposite sides of 1.

This is a genuine two-coordinate comparison at a single level, so it settles the question of principle: score is coordinate-dependent. The consequence for Problem 2.3 is that the true optimisation domain is not the (small, enumerable) set of levels but the *lattice of integral Möbius changes of t* , each trading singularity positions against denominator growth — one cannot move $|t'_c|$ outward for free, because rescaling t rescales A_n and B_n and hence the denominators. This is precisely Sol’s “extremal function” formulation, now with data attached. We return to it as the surviving open problem in §5.

4. S-1: THE LEVEL-10 BOUNDARY TEST

Problem 3.2 asks whether the knife-edge levels can be tipped. We report the level-10 computation in full, because every layer of the mechanism behaves *exactly* as the theory predicts — and the construction still fails, for a reason the first framing of S-1 had mis-stated.

4.1. The odd vector exists. The level-10 weight-4 oldspace contains the embeddings of $f_5 = \eta_1^4 \eta_5^4$, the level-5 newform. Its Atkin–Lehner eigenvalue is $\varepsilon_5 = +1$, read off from $a_5 = -5$; the general oldspace lemma (whose two previous instances are levels 12 and 6) gives $c = \varepsilon p^{k/2}$, here $c = 4$, so

$$(f_5|_4 W_{10}) = 4f_5(q^2),$$

and therefore

$$\boxed{f^- = f_5(q) - 4f_5(q^2)}$$

is the W_{10} -odd eigenvector. It vanishes at the fold automatically, by the same automorphy-factor parity argument as in §2. This is the *third* instance of the oldspace vanishing lemma and the exact structural analogue of level 12’s f^* . [VERIFIED]

4.2. The families are rectified. Level 10 supports integral principal solutions: for $F = 2E_2(q^2) - E_2(q)$ one gets $A_n = 1, 24, 168, 1752, \dots \in \mathbb{Z}$, and for $F = (5E_2(q^5) - E_2(q))/4$ one gets $1, 6, 54, 582, \dots \in \mathbb{Z}$. Nothing about level 10 is degenerate. [VERIFIED: exact, series to q^{36}]

4.3. Integrality holds. The odd-cusp companion $B_n = [t^n]F\theta_q^{-3}f^-$ satisfies

$$d_n^3 B_n \in \mathbb{Z} \quad [\text{VERIFIED: } n \leq 36, \text{ exact}].$$

The d_n^3 -integrality layer — which the programme has now observed at levels 6, 12 and 10, cuspidally and Eisenstein-wise, on several curves — holds yet again. (It remains *observed*, not proved, in the cuspidal case; see §6.)

4.4. And yet the construction fails. Two ways, both fatal:

- **The denominators grow.** $\log(\text{den } B_n)/n \approx 2.27\text{--}2.43$ for $n = 20\text{--}36$: slightly below the nominal d_n^3 -rate 3, but nowhere near bounded. There is no $d_n^{3-\varepsilon}$ miracle with ε big enough to matter, and certainly no boundedness.
- **The forms do not decay.** At $|t'_c| = 1.000$ the linear forms $A_n \xi - B_n$ are $O(1)$: they do not tend to 0 at all. A score of exactly 1 does not mean “arithmetic decides”; it means the analytic input is *zero*, and one would need bounded denominators *and* strictly decaying forms. Level 10 has neither.

Finding 4.1 (S-1 answered: NO). The cuspidal-Apéry cell is empty in the full scanned domain, now including the boundary levels. [VERIFIED: level 10 exact to q^{36} ; level 14 by the same unit-circle argument]

For level 14 we note a correction: its hauptmodul was mis-specified in the first run; the correct one is $(\eta_2 \eta_{14} / \eta_1 \eta_7)^3$, and its conjugate class sits at the same unit-circle scale, so the same argument applies.

Remark 4.2 (What survives of S-1). Only the normalisation-optimisation question of Lesson 3.3: whether some integral Möbius change of coordinate at *any* level can push $|t'_c|$ above the observed denominator rate. At the time this finding was recorded, the empirical rate for odd-cusp companions looked like $e^{2.3n}$ rather than e^{3n} , which would have lowered the bar — a level with $|t'_c| > e^{2.3} \approx 10$ and odd cusp forms is not obviously excluded. That deflation was real in the data and unexplained. Resolving it was posed as S-2, and it is the subject of the next section.

5. S-2: THE PRIME-LOCAL DENOMINATOR LAW, AND THE CLOSING OF THE LOOPHOLE

5.1. The question. Is the odd-cusp companion’s true denominator exponent $\kappa = 3$, or is it genuinely smaller? Everything hinges on this: the threshold to beat is e^κ , and the difference between $e^3 \approx 20.1$ and $e^{2.3} \approx 10.0$ is the difference between “no candidate levels” and “several”.

The protocol: compute B_n exactly for $n \leq 72$ for both integral choices of F at level 10, and split the denominator prime by prime. Write $h_p(n) = v_p(d_n) = \lfloor \log_p n \rfloor$ for the p -adic valuation of $\text{lcm}(1, \dots, n)$, and $e_p(n) = -v_p(B_n)$ for the power of p actually appearing in the denominator of B_n . The nominal law is $e_p \leq 3h_p$; the question is where it is sharp.

5.2. The law.

Finding 5.1 (Prime-local denominator law at level 10). Uniformly across both integral F ’s, for $n \leq 72$ [VERIFIED: exact rational arithmetic]:

$$e_p(n) = 3h_p(n) \quad \text{for } p \nmid 10 \quad (\text{generic primes: the full } d_n^3 \text{ cost}), \quad (3)$$

$$e_5(n) = 2h_5(n) \quad \text{exactly,} \quad (4)$$

$$e_2(n) \ll 2h_2(n) \quad (\text{the strongest deflation}). \quad (5)$$

Concretely for (3): $e_7 = 3$ at $n = 7$, $e_7 = 6$ at $n = 49$, and likewise for $p = 11, 13$. For (4): $e_5 = 2$ at $n = 5$, $e_5 = 4$ at $n = 25$. The auxiliary quantity $Q_n = \text{den}(B_n) / \gcd(\text{den}(B_n), d_n^2)$ contains every generic prime $\leq n$ to power exactly 1 — which is precisely the difference between $3h_p$ and $2h_p$ — and contains 5 never.

Two consequences, one negative and one positive.

Corollary 5.2 ($\kappa = 3$; the 2.3 was an artifact). *The globally fitted exponent 2.27 of §4 was a finite-size effect. It rises with n — reaching 2.75 by $n = 72$ — tracking $3\psi(n)/n$ minus $O(\log n)$ worth of level-prime savings, where ψ is Chebyshev’s function. The true exponent is $\boxed{\kappa = 3}$, and the analytic threshold stays e^3 .*

Hence the candidate universal “ d_n^2 -type” theorem — which would have made the boundary levels winnable — is [EXCLUDED] at generic primes. The scanner verdict of §3 stands, and now stands arithmetically as well as analytically: *the cuspidal-Apéry cell remains empty, with the arithmetic loophole closed.*

5.3. What is gained: level-prime deflation. The positive consequence is new arithmetic. The deflations at $p \mid N$ are real, structured, and provable-looking — they are the local behaviour of Frobenius on the Eichler extension at the primes of bad reduction. The 5-adic one turns out to have an exceptionally clean form. Probing $n \leq 60$ exactly:

Finding 5.3 (The exact 5-adic descent law). For the level-10 odd-cusp companion,

$$\boxed{v_5(B_{5m}) = v_5(B_m) - 2}$$

[VERIFIED: all tested $m \leq 11$, uniformly, including cases with $v_5(B_m) < 0$]. That is: each 5-adic digit of n costs exactly *two* powers of 5. The mechanism visible in the data: the division by m^3 in the recurrence loses three powers, and the numerator of the forced recurrence recovers exactly one.

Everything else about the 5-adic side follows from Finding 5.3 by digit bookkeeping: the equality set of $e_5 = 2h_5$ (which holds for 34 of the 60 values $n \leq 60$) and its exception windows — the 5-adic digit patterns 21_5 – 24_5 , 41_5 – 44_5 , 111_5 – 114_5 , ... — are exactly what the descent law predicts. Moreover $25 \cdot B_{5m} \bmod 5$ is a unit whose value tracks the pair (A, B) : this is the digit-transition constant of a mixed Lucas congruence, one experiment away from explicit.

Note the shape of the law: it is an *equality*, a 5-adic recursion, not an inequality. That is unusual and is what makes it look provable.

Problem 5.4 (S-3, open). Prove $v_5(B_{5m}) = v_5(B_m) - 2$ for all m . Two routes are identified: the source side (the action of U_5 on $\theta_q^{-3}f^-$, which would explain why the law is independent of the choice of F — the observed F -invariance is otherwise unexplained), or the forced recurrence directly (shortest). [OPEN]

5.4. The surviving extremal problem. With the arithmetic loophole closed, exactly one route to a cuspidal irrationality proof survives inside this mechanism, and it is the one Lesson 3.3 opened.

Problem 5.5 (The integral-model extremal problem).

Find a quadruple (N, t, F, f^-) — a level, an *integral* coordinate t on the modular curve, a rectified weight-2 family F with integral t -coefficients, and a W_N -odd cusp form f^- of weight 4 — such that the conjugate singular value satisfies

$$|t'_c| > e^3 \approx 20.09$$

in that coordinate, with the d_n^3 -integrality of the companion preserved. [OPEN]

Equivalently: optimise the score over the lattice of integral Möbius changes of coordinate, where every outward push of $|t'_c|$ is paid for in denominator growth.

This is a genuine extremal problem, not a restatement of Problem 2.3: the scan fixed one standard coordinate per level, and Lesson 3.3 proves that this choice can decide the race. Level 5

is the existence proof that the coordinate lattice matters — there the asymmetric normalisation gains a factor $\varphi^{10} \approx 123$ over the symmetric one.

Together with Problem 5.4, this is what the arc leaves open. Everything else in the cell is settled, for the scanned class: analytically by Table 2, at the boundary by Finding 4.1, and arithmetically by Finding 5.1.

statement	grade
$\Phi_\gamma(\tau_c) = 0$ (Apéry’s source is odd, vanishes at the fold)	[VERIFIED: 10^{-122}] + [PROVED] parity argument
$(f _k W_{Np}) = \varepsilon p^{k/2} f(q^p)$ on the oldspace (three instances)	[PROVED] (level 12), [VERIFIED] (levels 6, 10)
Level 6 is the unique $r = 3$ winner, $N \leq 25$, standard hauptmoduln	[VERIFIED: Table 2]
$S_4^-(\Gamma_0(6)) = 0$	[PROVED] ($\dim S_4 = 1$, f_6 Fricke-even)
Boundary levels 10, 14 do not win	[VERIFIED: $n \leq 36$ exact]
$d_n^3 B_n \in \mathbb{Z}$ for cuspidal companions	[VERIFIED: $n \leq 36$ –40] — <i>not</i> proved
$e_p = 3h_p$ at generic p , i.e. $\kappa = 3$	[VERIFIED: $n \leq 72$, both F]
Universal d_n^2 -type theorem	[EXCLUDED: generic primes]
$v_5(B_{5m}) = v_5(B_m) - 2$	[VERIFIED: $m \leq 11$] — [OPEN] as S-3
Integral-model extremal problem	[OPEN]
Catalan / Koutschan–Zudilin outside the modular class	[EXCLUDED: bounded scans]

TABLE 3. Evidence ledger for the claims of this paper.

5.5. Summary of the position.

6. HONEST LIMITS

This section exists because the central result is a negative, and a negative is only as good as its stated domain. Everything below is a limitation of the present work, not a hedge.

6.1. Search bounds.

- (L1) **Levels ≤ 25 .** Larger levels were not scanned. There is no theoretical bar at 25; it is where genus-zero standard eta hauptmoduln run out in the tabulated list used.
- (L2) **One hauptmodul per level.** Exactly one standard integral eta-quotient coordinate was scanned per level — with the sole exception of level 5, whose two normalisations straddle the win/no-gain line. That exception is not a nuisance: it is the *proof* that the single-coordinate scan is incomplete in principle (Lesson 3.3), and it is why Problem 5.5 is open rather than closed.
- (L3) **Seeded zero-finding.** The Newton search for critical values is seeded. At levels 6, 12 and 18 the second class was taken from the known family operators or from scale estimates rather than found fresh. For levels 6 and 12 these values are independently known and exact; for level 18 only the *scale* is used, which suffices to rule out a win but does not resolve the second class.

- (L4) **Odd-part dimensions not split in general.** The scan reports $\dim S_4(\Gamma_0(N))$, not $\dim S_4^\pm$. The splitting was carried out only where it mattered — level 6 (where $\dim S_4 = 1$ and the sign is $+$) and level 10. At other losing levels the odd dimension is irrelevant, since they lose the race regardless.
- (L5) **Weight $r = 2$ under-scanned.** The weight-3-with-character spaces relevant to $r = 2$ were not systematically scanned; the reported $r = 2$ column is a by-product of the same fold data.
- (L6) **Level 14's coordinate.** Mis-specified in the first run and corrected afterwards (see §4). The corrected level sits at the same unit-circle scale, so the verdict is unchanged — but the incident is a reminder that hauptmodul bookkeeping is error-prone and every entry of Table 2 should be re-derived by any reader who wants to build on it.

6.2. What is assumed rather than proved.

- (A1) **The d_n^r cost.** The identification $\delta = e^r$ assumes that the cuspidal companion's denominators really are governed by d_n^r . For the Eisenstein (classical) cases this is proved. For the cuspidal companions it is *observed*: exactly to $n \leq 40$ at levels 12 and 6, to $n \leq 36$ at level 10, and prime-locally to $n \leq 72$. It is not a theorem. A reader looking for a way through Problem 2.3 should note that this cuts both ways: the cuspidal denominators could in principle be *worse* than d_n^r at some level, which would only strengthen the negative — or better at some unexamined level, which would weaken it.
- (A2) **The rate dictionary.** The identifications $\alpha = 1/|t_c|$, $|\beta| = 1/|t'_c|$ are the standard singularity analysis of the Picard–Fuchs operator, valid when the source vanishes at the fold to the required order and the fold is a regular singular point with the expected exponents. They were checked against the two classical constructions and the manufactured level-12 apparatus. They are not re-derived here for each scanned level.
- (A3) **Numerical fold data.** Table 2's entries are numerical, at 50 digits working precision, with a per-level Möbius residual check. The exact values are known independently only at levels 5, 6, 8, 12 and 20.
- (A4) $\Phi_\gamma(\tau_c) = 0$. The numerical statement is overwhelming (10^{-122}), and the parity argument that explains it is a proof *given* the Atkin–Lehner eigenvalue of Φ_γ , which is read off the Eisenstein decomposition rather than derived here from first principles. We grade the combination [PROVED] but flag the input.
- (A5) **Mechanism claims.** The two effects of fold-vanishing (§2(i),(ii)) are argued and observed; the period-parity half in particular has a written route with one normalisation constant still open in the source ledger. Nothing in this paper's verdict depends on that constant, since the verdict is about rates.

6.3. What the negative does and does not say. It says: within an explicit finite family of modular constructions — one standard integral coordinate per genus-zero level $N \leq 25$, odd sources, Atkin–Lehner folds — Apéry's level-6 Eisenstein construction is the only one whose score is below 1, and no level with a nonzero odd cusp space comes within a factor of 4.

It does not say that no cuspidal irrationality proof exists, nor even that none exists by this mechanism. Two doors are explicitly open: Problem 5.5 (a different integral coordinate, at any level, possibly large), and the possibility of a structurally different mechanism altogether — for which the two hypergeometric control experiments of §2 suggest the known candidates in the literature are not it.

7. CLOSING: WHAT A SCAN IS FOR

A scan of this kind produces three kinds of output, and it is worth separating them.

A verdict. The cuspidal-Apéry cell is empty in the scanned domain, analytically and arithmetically. If one is looking for a second Apéry, one should not look here.

A reframing. Before the scan, “why is Apéry’s source Eisenstein?” had no answer beyond “that is what the computation gives”. After Finding 2.1 and Observation 2.2, it has one: Apéry’s construction is the odd-source-vanishing-at-the-fold mechanism, and at the only level where that mechanism wins its race, the odd part of the weight-4 space is purely Eisenstein. The Eisenstein-ness is forced. Table 1 is the whole story, and three of its four cells are now understood.

New arithmetic that nobody was looking for. The scan was a search for a race winner and found none; but en route it produced the third instance of the oldspace vanishing lemma, the $\kappa = 3$ generic denominator law, the exclusion of the universal d_n^2 theorem, and the exact 5-adic descent law $v_5(B_{5m}) = v_5(B_m) - 2$ — an equality, not an inequality, with two identified proof routes. In this programme bounded negatives are deliverables, and the structure they force out on the way is usually the durable part.

What remains, precisely: prove the descent law (Problem 5.4), and solve the extremal problem over integral models (Problem 5.5). Failing both, a cuspidal irrationality proof needs a mechanism this paper has not scanned.

Acknowledgements. The score formulation of the race, the scanner proposal, the protocol for the prime-local denominator split, and the extremal-function framing of the surviving open problem are all due to our collaborator Sol.

APPENDIX A. PROVENANCE

For a reader wishing to reproduce or extend the computations, the underlying records of this programme are:

- `work/SCANNER_LEDGER.md` — the scan itself: method (§1), the table (§2), the verdict (§3), the normalisation lesson (§4), honest limits (§5), the level-10 boundary test (§6), the prime-local law (§7), and the 5-adic descent law (§8). All of §3–§6 above tracks this ledger.
- `work/D1_CUSPIDAL_APPARATUS.md` — the manufactured cuspidal apparatus, and (§9) the race framework together with the $\Phi_\gamma(\tau_c) = 0$ discovery and the 2×2 table.
- `work/CATALAN_CONTROL.md`, `work/KZ_ELLIPTIC_REPORT.md` — the hypergeometric boundary: why the modular class is the arena.
- `work/z5eps/eps65_scanner.py` — the scanner: eta hauptmodul table, numerical Atkin–Lehner Möbius fit, fixed points, $\dim S_k(\Gamma_0(N))$, and the score columns for $r = 2, 3$. The remaining computations (levels 10, 14; the prime-local splits; the descent law probe) were run inline in the same session’s exact-arithmetic environment.

The companion papers of this arc are the cuspidal-companion paper (the $L(f_6, 3)/2$ apparatus, the oldspace lemma, the parity mechanism and the six-item proof package) and the modular-anchor papers of the wider programme, to which we refer for the theory of sources $\Phi = t\sigma^r/(PF)$ and the fold connection formula.

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